

Wheat Head Detection

For our final project in APS360, my group decided to tackle wheat head detection using the GWHD dataset. The dataset consisted of photos of wheat fields and bounding boxes for wheat heads.

As a baseline we used just a CNN, but we also tested two other architectures: contour + CNN and autoencoder + CNN. Contour involved using that OpenCV filter to enhance the wheat field images before feeding into the CNN, whereas the autoencoder was used to generate a mask of the wheat heads before feeding back into the CNN.

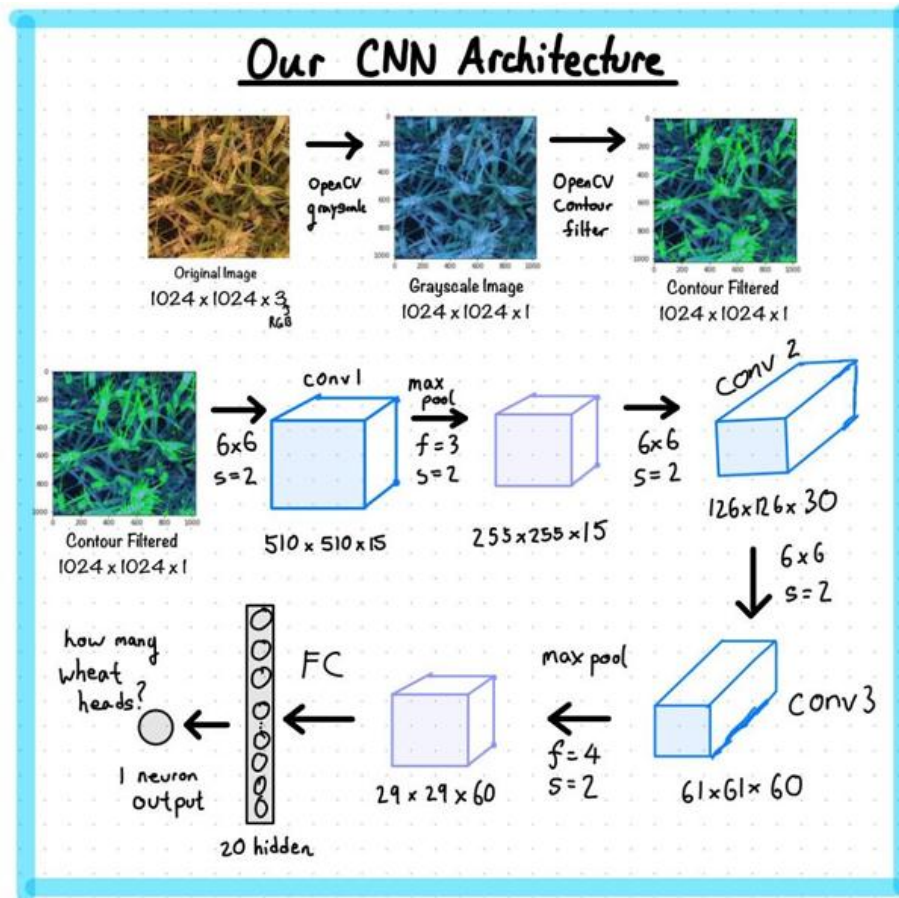


Figure 1. Contour + CNN

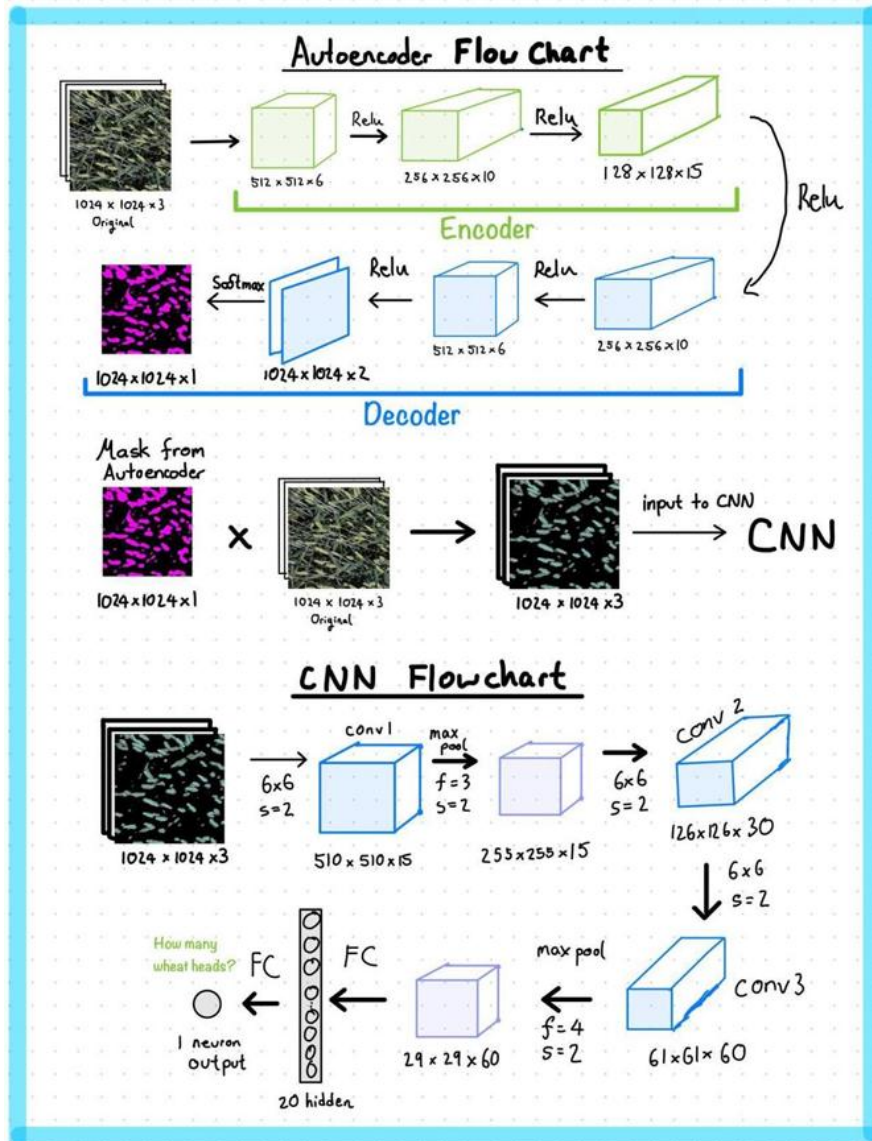


Figure 2. Autoencoder + CNN

The models were evaluated by looking at a normal distribution of the RMS error, and as expected, both enhanced models performed better than the baseline. With new data, however, contour performed significantly better.

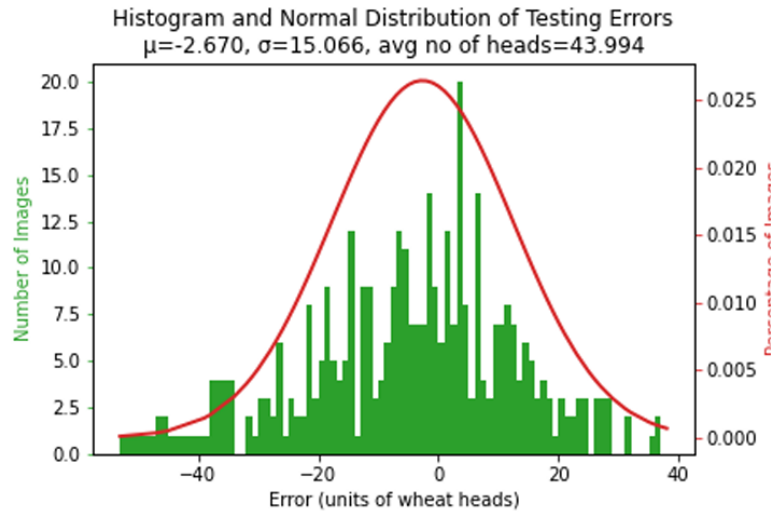


Figure 3. AutoEnc + CNN Testing

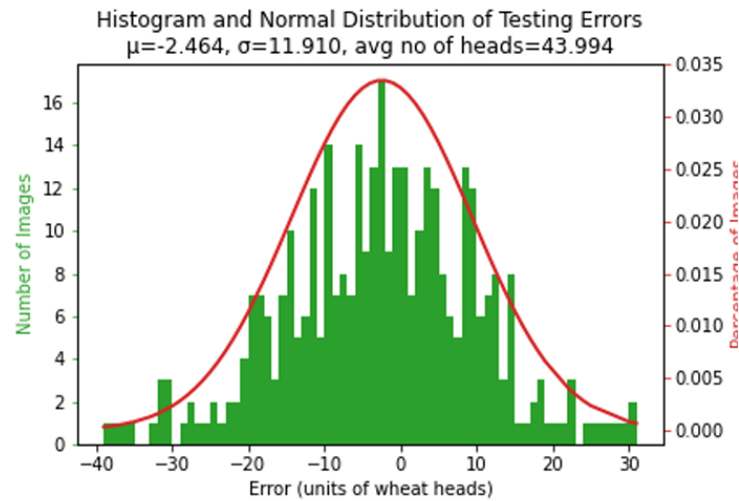


Figure 4. Contour + CNN Testing

Github: <https://github.com/CarbonicKevin/2020-Summer-APS360-Project>